

**B.Tech. Degree III Semester Regular/Supplementary Examination
November 2023**

**CS 19-202-0302 LOGIC DESIGN
(2019 Scheme)**

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Understand about the basic number systems and the conversion between them.
 CO2: Manipulate boolean expressions and simplify them.
 CO3: Design combinational circuits for any given problem.
 CO4: Design sequential circuits, flip-flops etc.
 CO5: Design circuits like counters, registers etc.
 CO6: Familiarize with the basic principles of memory, design of memory etc.
 CO7: Gain knowledge about the basics of integrated circuits.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A(Answer **ALL** questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	Given the two octal numbers X = 13456 and Y = 10321, perform the subtraction. (i) X - Y and (ii) Y - X by using 8's complements.	3	L2	1	1,2,4
	(b)	Differentiate between canonical and standard forms of Boolean functions. Express the Boolean function $F(x,y,z) = (xy+yz)(y+xz)$ as a sum of minterms.	3	L2	2	1,2,4
	(c)	Design a 4 x 16 decoder using two 3 x 8 decoders.	3	L3	3	1,2,4
	(d)	Design a 3 bit magnitude comparator.	3	L4	3	1,2,3,4,12
	(e)	Design a 4 bit ring counter. How does it differ from a Johnson counter?	3	L3	5	1,2,4
	(f)	What are the basic differences between FPGA, PLA and PAL?	3	L1	6	1,2,4
	(g)	Define the following terms: (i) Fan out (ii) Propagation delay (iii) Noise margin.	3	L1	7	1,2,4
	(h)	Write notes on Programmable logic array.	3	L4	6	1,2,4

PART B

(4 × 12 = 48)

- II. (a) Simplify the following expression using the K-Map method by finding essential prime implicants.
 $F(A, B, C, D) = \Sigma(4, 2, 7, 10, 12)$
 $d(A, B, C, D) = \Sigma(0, 6, 8)$
- (b) Find the complement of $F = wx + yz$; then show that $FF^1 = 0$ and $F + F^1 = 1$.

OR

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		Marks	BL	CO	PO
III.	(a) Obtain the truth table of the following function, and express the function in sum-of-min-terms and product-of-maxterms form: $F = bd^1 + acd^1 + ab^1c + a^1c^1$	6	L2	2	1,2,4
	(b) Simplify the following Boolean function using the Quine McCluskey method. $F(A, B, C, D) = \sum(0, 2, 3, 5, 7, 8, 10, 11, 14, 15)$	6	L3	2	1,2,4,12
IV.	(a) Design a combinational circuit with three inputs and one output. (i) The output is 1 when the binary value of the inputs is less than 3. The output is 0 otherwise. (ii) The output is 1 when the binary value of the inputs is an even number.	6	L4	3	1,2,12
	(b) (i) Design a half-subtractor circuit with inputs x and y and outputs Diff and B_{out} . The circuit subtracts the bits $x - y$ and places the difference in Diff and the borrow in B_{out} . (ii) Design a full-subtractor circuit with three inputs x, y, B_{in} and two outputs Diff and B_{out} . The circuit subtracts $x - y - B_{in}$, where B_{in} is the input borrow, B_{out} is the output borrow, and Diff is the difference.	6	L3	3	1,2,12
OR					
V.	(a) What is Carry Propagation delay? Design a 4-bit binary parallel adder with look ahead carry generator.	6	L4	3	1,2,4,12
	(b) Implement the following function using a multiplexer. $F(A, B, C, D) = \sum(0, 2, 5, 8, 10, 14)$	6	L3	3	1,2,4,12
VI.	(a) Show that the characteristic equation for the complement output of a JK flip-flop is $Q^1(t+1) = J^1Q^1 + KQ$	6	L4	4	1,2,3,12
	(b) Design a BCD ripple counter using JK flip-flops.	6	L3	5	1,2,4,12
OR					
VII.	(a) Design a 4 bit synchronous up counter using JK flipflops.	6	L3	5	1,2,4,12
	(b) Explain the differences among a truth table, a state table, a characteristic table, and an excitation table. Also, explain the difference among a Boolean equation, a state equation, and a characteristic equation.	6	L4	4	1,2,4,12
VIII.	(a) Design a PLA circuit to implement the functions $F1 = \sum(0, 1, 2, 4)$ $F2 = \sum(0, 5, 6, 7)$	6	L3	6,7	1,2,4,12
	(b) What is meant by coincident decoding? With a neat diagram, illustrate the two dimensional decoding structure for a 1K-word memory.	6	L4	6	1,2,4
OR					
IX.	Draw and explain the circuit for a TTL gate with Totem pole output driver.	12	L2	7	1,2

Bloom's Taxonomy Levels

L1 = 8%, L2 = 17%, L3 = 42%, L4 = 33%.
